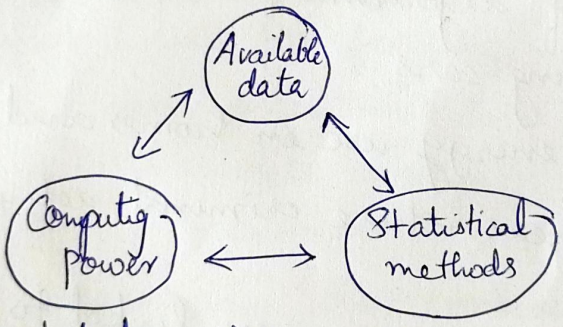
Module 2

Machine Learning

- The field of machine learning provides a set of algorithms that transform data into actionable knowledge.
- The field of study interested in the development of computer algorithms to transform data into intelligent action is known as Machine Learning
- This field originated in an environment where available data, statistical methods, and computing power rapidly and simultaneously evolved.



A closely related sibling of ML → data mining

eg: Recommendation Engines (Netflix)

- Machine learning is used in internet search engines, email filters to sort out spam, websites to make personalised recommendations, banking software to detect unusual transactions, and lots of apps on our phones such as voice recognition.

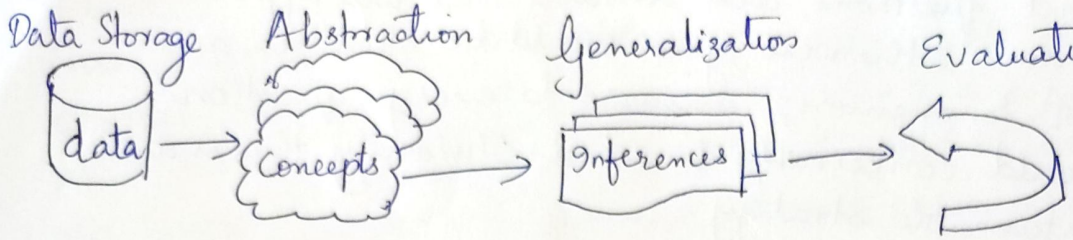
applications of Machine learning includes.

- Identification of unwanted spam messages in e-mail
- Segmentation of customer behavior for targeted advertising
- Forecasts of weather behaviour and long-term climate changes.
- Reduction of fraudulent credit card transactions.
- Actuarial estimates of financial damage of storms and natural disasters.
- prediction of popular election outcomes
- Development of algorithms for auto-piloting drones and self-driving cars.
- Optimization of energy use in homes and office buildings
- projection of areas where criminal activity is most likely
- Discovery of genetic sequences linked to diseases.

How machines learn

Regardless of whether the learner is a human or machine, the basic learning process is similar. It can be divided into Four interrelated components:

1. Data storage
2. Abstraction
3. Generalization
4. Evaluation



The fig illustrates the steps in the learning process.

with computers these processes are explicit, and because of the entire process is transparent, the learned knowledge can be examined, transferred, and utilized for future action.

① Data Storage

- Utilizes Observation, memory, and recall to provide a factual basis for further reasoning.
- All learning must begin with data. Humans and computers alike utilize data storage as a foundation for more advanced reasoning.
- In a human being, this consists of a brain that uses electrochemical signals in a network of biological cells to store and process observations for short and long-term recall. computers have similar capabilities of short- and long-term recall using hard disk drives, flash memory and Random access memory (RAM) in combination with a central processing unit (CPU).

To better understand the nuances of this idea, it may help to think about the last time you studied for a difficult test, perhaps for a university final exam or a career certification. Did you wish for an eidetic (photographic) memory? If so, you may be disappointed to learn that perfect recall is unlikely to be of much assistance. Even if you could memorize material perfectly, your rote learning is of no use, unless you know in advance

the exact questions and answers that will appear in the exam. otherwise, you would be stuck in an attempt to memorize answers to every question that could conceivably be asked. Obviously this is an unsustainable strategy.

Instead, a better approach is to spend time selectively, memorizing a small set of representative ideas while developing strategies on how the ideas relate and how to use the stored information. In this way, large ideas can be understood without needing to memorize them by rote.

② Abstraction

- involves the translation of stored data into broader representations and concepts.
- This works of assigning meaning to stored data occurs during the abstraction process, in which raw data comes to have a more abstract meaning.
- This type of connection say between an Object and its representation, is exemplified by the famous Rene Magritte painting 'The Treachery of Images':



Ceci n'est pas une pipe. ⇒ caption

The painting depicts a tobacco pipe with the caption ("this is not a pipe"). The point Magritte was illustrating is that a representation of a pipe is not truly a pipe. Yet, in spite of the fact that the pipe is not real, anybody viewing the painting easily recognizes it as a pipe.

his suggests that the observers mind is able to connect the picture of a pipe to the idea of a pipe, to a memory of a physical pipe that could be held in the hand. Abstracted connections like these are the basis of knowledge representation, the formation of logical structures that assist in turning raw sensory information into a meaningful insight.

- During a machine's process of knowledge representation, the computer summarizes stored raw data using a model, an explicit description of the patterns within the data.

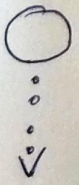
There are many different types of models.

- Mathematical equations
- Relational diagrams such as trees and graphs
- Logical if/else rules
- Groupings of data known as clusters.

- The process of fitting a model to a dataset is known as training. When the model has been trained, the data is transformed into an abstract form that summarizes the original information.

For instance:- Discovery of gravity — By fitting equations to observational data, Sir Issac Newton inferred the concept of gravity.

Observations ⟶ Data ⟶ Model

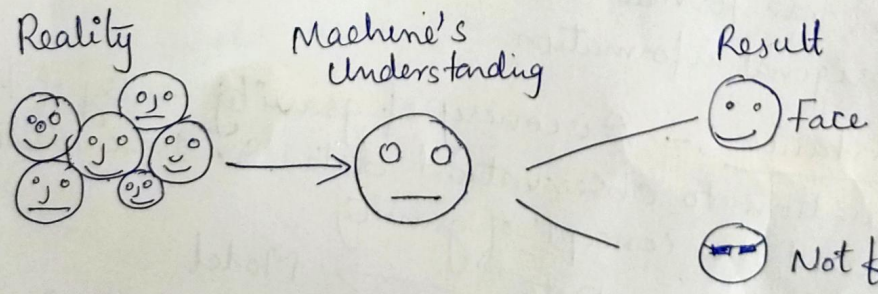


Distance	Time
4.9 m	1s
19.6 m	2s
44.1 m	3s
78.5 m	4s

$g = 9.8\ m/s^2$

③ Generalization : uses abstracted data to create knowledge and inferences that drive action in new contexts.

- The term generalization describes the process of turning abstracted knowledge into a form that can be utilized for future action, on tasks that are similar, but not identical, to those it has seen before.
- Traditionally, it has been imagineed as a search through the entire set of models (i.es theories or inferences) that could be abstracted during training
- In other words, if you can imagine a hypothetical set containing every possible theory that could be established from the data, generalization involves the reduction of this set into a manageable number of important findings.
- Machine learning algorithms generally employ shortcuts that reduce the search space more quickly. toward this end, the algorithm will employ heuristics, which are educated guesses about where to find the most useful inferences.



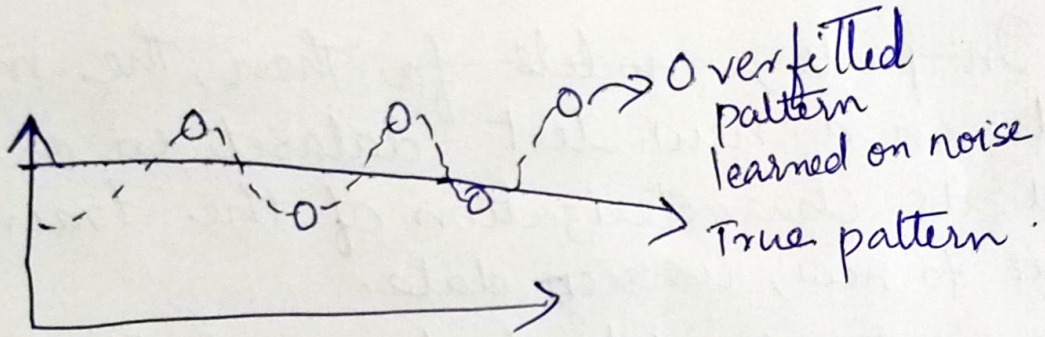
Evaluation

- provides a feedback mechanism to measure the utility of learned knowledge and inform potential improvements.
- Generally, evaluation occurs after a model has been trained on an initial training dataset.
- In parts, ~~models fa~~ then, the model is evaluated on a new test dataset in order to judge how well its characterization of the training data generalizes to new, unseen data.
- In parts, models fail to perfectly generalize due to the problem of noise, a term that describes unexplained or unexplainable variations in data. Noisy data is caused by seeming random events.
  1. Measurement error due to imprecise sensors that sometimes add or subtract a bit from the readings.
  2. Issues with human subjects, such as survey respondents reporting random answers to survey questions, in order to finish more quickly.
  3. Data quality problems, including missing, null, truncated, incorrectly coded or corrupted values.
  4. Phenomena that are so complex or so little understood that they impact the data in ways that appear to be unsystematic.

Trying to model noise is the basis of a problem called overfitting. Because most noisy data is unexplainable by definition, attempting to explain the noise will result in erroneous conclusions that do not generalize well to new cases.

Efforts to explain the noise will also result in more complex models that will miss the true pattern that the learner tries to identify.

A model that seems to perform well during training but does poorly during evaluation, is said to be overfitted to the training dataset, as it does not generalize well to the test dataset.



Machine learning in practice

(1) Data collection : The data collection step involves gathering the learning material, an algorithm will use to generate actionable knowledge.

In most cases, the data will need to be combined into a single source like a textfile, spreadsheet, or database.

(2) Data exploration and preparation — The quality of any ML project is based largely on the quality of its i/p data. Thus, it is important to learn more about the data and its nuances during a practice called data exploration.

Additional work is needed to prepare the data for the learning process. This involves.

fixing or cleaning so-called "messy" data,
eliminating Unnecessary data.
recoding the data to conform to the learner's expected i/Ps.

(3) Model training : By the time the data has been prepared for analysis, you are likely to have a sense of what you are capable of learning from the data.

The specific ML task chosen will inform the selection of appropriate algorithm, and the algorithm will represent the data in the form of model.

(4) Model evaluation :- it is important to evaluate how well the algorithm learns from its experience. Depending on the type of model used, you might be able to evaluate the accuracy of the model using a test dataset

(5) Model improvement :- If better performance is needed, it becomes necessary to utilize more advanced strategies to augment the performance of the model.